

General Instruction

1. Student Instruction

Basic Rules

1. Each team must consist of 1–4 students.
2. Every team must submit via GitHub Repository Link
 - Working Project (Frontend + Backend)
 - Final Presentation (PPT and Demo)
3. Project must be:
 - Original
 - Developed during the hackathon timeline
4. AI tools (ChatGPT, Copilot, etc.) are allowed for guidance, but:
 - You must understand your code.
 - You must explain your logic during judging.

Timeline Discipline

- No late submissions.
- Final Git commit time will be verified.
- Last-minute ZIP submissions without proper repository history will be rejected.

Project Expectations

Your project must include:

- Working UI
- Proper Backend/API
- Database integration
- Deployment (recommended and Preferable)

2. Git Repository & Team Development Rules

Mandatory Requirements

1. Repository must be created by the team leader within 24 hours of releasing the problem statement.
2. Repository name format: NavKalpana-TeamCode
Example: NavKalpana-RICR-NK-0001
3. All team members must:
 - Must be the contributor in repo
 - Use individual GitHub accounts
 - Contribute via commits
 - Avoid single-person commits for team projects

Commit Rules

- Minimum 10 meaningful commits.
- Avoid a single bulk commit at the end.
- Commit messages must be clear.
 - Good Commit Example:
 - Added login authentication
 - Integrated payment API
 - Fixed cart validation bug

- Bad Commit Example:

- Update
- Final
- Added
- done

Bad Commit Messages will lead to negative marking

Branching Rule (Recommended)

- main – Final stable version
- All the Members must Create Separate Branches which further merges to the Main Branch
 - Example:
 - feature/login
 - feature/register
 - bug/update etc.

3. Recommended Project Folder Structure

Repository Name Example: *NavKalpana-RICR-NK-0001*

Standard Structure

- ❖ NavKalpana-TeamCode
 - frontend
 - backend
 - docs
 - problem-statement.pdf
 - architecture-diagram.png
 - api-documentation.md
 - presentation.pptx
 - README.md

4. README.md Must Include

- Project Title
- Team Members & Roles
- Problem Statement
- Tech Stack Used
- Installation Steps
- API Endpoints
- Screenshots
- Future Improvements

5. What Judges Will Verify

Technical Verification

- Git commit history
- Contribution by all members
- Code understanding

- Folder structure clarity
- Backend API functionality
- Database integration
- Error handling

Innovation Check

Judges will evaluate:

- Whether it solves a real problem
- Practical usability
- Scalability potential
- Meaningful use of AI (if included)

Viva Round Questions

Each member may be asked:

- Explain your module.
- Explain database schema.
- Explain API flow.
- Explain deployment steps.
- Explain one technical challenge you faced.

6. Important Warning

Immediate disqualification may occur if:

- Code is copied from the internet without modification
- Purchased templates are used without proper logic implementation
- Repository is created only on the final day
- Only UI is presented without functionality
- Fake data is shown without backend logic (unless clearly defined prototype)
- Only one member answers all questions

7. Bonus Considerations

- Clean UI/UX
- Proper Authentication
- Role-based access control
- AI/ML Integration (if required)
- Deployment (Vercel, Render, AWS, Azure)
- Clean code practices

Final Note

The hackathon is not only about coding. It is about teamwork, structured thinking, problem solving, and clear communication. Build a meaningful solution and be prepared to explain it confidently.

FitAI

Adaptive Fitness Intelligence Platform

1. Executive Summary

FitAI is a web-based adaptive fitness planning and coaching platform that generates personalized workout and diet plans, monitors adherence, analyses behavioural patterns, intelligently adjusts training and nutrition, and provides AI-assisted guidance.

The system operates as a closed-loop adaptive engine:

Profile → Plan → Execute → Track → Analyze → Adjust → Coach → Repeat

FitAI is not a static template generator.

It is a habit-driven adaptive fitness intelligence system.

2. Product Vision

To create a structured, safe, adaptive digital fitness companion that:

- Builds long-term fitness habits
- Adjusts plans intelligently
- Prevents overtraining
- Tracks measurable transformation
- Uses explainable AI for coaching
- Encourages sustainable health improvement

3. Target Users

Primary Users:

- Fitness beginners
- Weight loss seekers
- Muscle gain beginners/intermediate
- Users seeking structure without hiring a trainer

Secondary Users:

- Admin (template management, analytics oversight)

4. System Architecture

FitAI consists of two major layers:

PART 1 – Core Adaptive Fitness Engine (~75%)

PART 2 – Advanced Intelligence & Premium Enhancements (~25%)

PART 1 – CORE ADAPTIVE FITNESS ENGINE

This layer delivers the full adaptive loop.

5. Authentication & Security

Email/password login

Secure password hashing (bcrypt / Argon2)

JWT session management

Session timeout

Clear medical disclaimer on every plan page
Safe calorie floor enforcement

6. Health & Goal Profile Setup

Required Inputs

Age

Biological sex

Height

Current weight

Activity level

Experience level

Primary goal:

- Weight Loss
- Muscle Gain
- Body Recomposition
- Maintain
- Improve Endurance

Calculated Values

BMI

Maintenance calories (Mifflin-St Jeor formula)

Suggested calorie target

Calorie Safety Floors:

Women → Minimum 1200–1400 kcal

Men → Minimum 1500–1800 kcal

7. Weekly Workout Plan Generator

Input Factors

Goal

Experience level

Activity level

Output Structure (7 Days)

Day name + Focus

Exercises per day

Sets × Reps

Rest intervals

Brief form guidance

Safe progressive structure:

Beginner → Lower volume

Intermediate → Moderate volume

8. Weekly Diet Plan Generator

Inputs

Calorie target
Goal (deficit / surplus / maintenance)
Macro template

Output Structure

Meals per day
Calories per meal
Macro split
Daily total

Example Macro Split:

Weight Loss → 40% Protein / 30% Carbs / 30% Fat

Muscle Gain → 30% Protein / 50% Carbs / 20% Fat

9. Progress Tracking Module

Users log:
Weekly weight
Workout completion (Completed / Partial / Skipped)
Diet adherence (Followed / Mostly / Deviated)

Visualizations

Weight trend (4–12 weeks)
Workout completion %
Diet adherence %

10. Body Measurement Tracking

Tracks:
Waist
Chest
Hips
Arms
Thighs
Displays:
Measurement change since start
Monthly comparison
Line graph visualization
Purpose:
Weight alone is insufficient for muscle gain or recomposition.

11. Habit Intelligence Engine

Habit Score Formula

Habit Score =
 $(\text{Workout Adherence} \times 0.60) +$
 $(\text{Diet Adherence} \times 0.40)$
Range: 0–100

Displayed

Weekly Habit Score
Monthly Habit Average
Current Streak

Drop-off Risk Detection

Triggers:

3 consecutive missed workouts
No log for 14 days
Diet adherence < 40% for 2 weeks

System Response:

Motivational prompt
Offer lighter plan
Offer schedule reset

12. Energy & Recovery Intelligence

Daily check-in:

Energized
Normal
Slightly Fatigued
Very Tired

Adjustment Logic

If fatigue high:
Reduce intensity
Lower volume
Swap heavy workout for mobility
If 3 fatigue flags in 7 days:
Force recovery day

13. Progressive Overload Engine

Scientific training principle implementation.

Adjustment Rules

If workout completion $\geq 90\%$ for 2 weeks:
Increase volume by 5–10%
If completion < 50%:
Reduce intensity
If goal = Muscle Gain:
Increase load gradually
If goal = Weight Loss:
Increase cardio gradually

14. Goal Timeline Forecast Engine

Uses average weekly weight change.

Forecast Formula:

Estimated Weeks to Goal =

$(\text{Goal Weight} - \text{Current Weight}) \div \text{Avg Weekly Change}$

Displayed:

Estimated completion date range

Confidence band

Updates dynamically.

15. AI Chat Fitness Assistant

Interactive coaching module.

User asks:

Why am I not losing weight?

Should I increase protein?

Can I skip cardio?

Why am I fatigued?

AI Logic (Phase 1)

Rule-based system using:

Weight trend

Habit score

Calorie deficit

Fatigue logs

Workout adherence

Output

Data-backed explanation

1–2 actionable steps

Safety disclaimer

Example:

“You are losing 0.1 kg/week. Increasing the deficit by 150 kcal may help.”

Future upgrade:

LLM-powered contextual coaching.